S475/1
SUBSIDIARY
MATHEMATICS
Paper 1
July 2026
$2\frac{1}{2}$ hours

ACEITEKA JOINT MOCK EXAMINATIONS 2026

Uganda Advanced Certificate of Education

SUBSIDIARY MATHEMATICS

PAPER 1

Time: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of two sections; A and B.

*It has **six** examination items.*

*Section A has **two** compulsory items.*

*Section B has two parts; **I** and **II**. Respond to only one item from **each** part.*

All responses must be written in the answer booklet(s) provided.

Graph paper is provided.

Use silent, non-programmable scientific calculators and mathematical tables.

SECTION A

ITEM 1

A civil engineering team in **a certain District** is installing an underground water pipeline in a newly planned residential area.

The two main water source are located at point P(3,1) and Q(-1,-3)

To regulate water pressure , the engineers plan to install a monitoring device at the midpoint of the two main water source.

However, for safety and efficiency:

- The device must be connected to a control centre at O(0,0)) using a cable.
- Engineers are concerned about the angle between the pipeline direction connecting the water source Q and the device.

If the angle is more than the acute, signal transmission is considered inefficient

Task:

(a) Use mathematical justification, advise the engineers

ITEM 2

A farmer visited an agricultural exhibition and learned about a modern irrigation system designed to address the challenge of insufficient water supply .
The irrigation system supplies water to crops at a rate modeled by:

$R(t) = 12t - 3t^2$ litres per hour,
where t is the time in hours and R(t) is the rate of water supply.

For successful irrigation, the maximum rate at which water is supplied should exceed 10 litres per hour.
However, the amount of water stored in the reservoir at the farmer's farm decreases over time due to leakage from the storage tank. The rate of change of the amount of water in the tank is proportional to the amount of water remaining .

Initially, the tank contained 1,000 litres of water. After one hour, the amount of water in the tank had decreased by 20 litres due to leakage. The farmer wishes to estimate the amount of water that will remain in the tank after 3 hours.

If the amount of water remaining after 3 hours is less than 950 litres, the farmer will reduce the opening of the tank tap to minimize further water loss.

Task:

Use mathematical calculations to advise the farmer.

SECTION B

This section has two parts, I and II

PART I

Respond to only one item in this part.

ITEM 3

A group of five Senior Two students in a certain school developed a natural hair product as their Biology project. However, they lacked sufficient knowledge to estimate the shelf life of the product before it expires. Some Senior Six students tested the product and advised that its shelf life is modeled by the equation:

$3^{2x} - 4(3^x) = -3$

where x represents the shelf life of the product in years.

The group plans to produce at least 300 pieces of the product if the shelf life exceeds 2 years.

The group was later allowed to supply the product to the school canteen. The demand for the product increased by a constant number of pieces each week. Records showed that 27 pieces were supplied in the seventh week and 47 pieces in the twelfth week. The group intends to undertake mass production to satisfy the demand up to the thirty-sixth week. However, if the total number of products required exceeds 2,630, the group will recruit additional workers to support production and supply.

The school administration also advised the group to form a gender-balanced committee of three members to promote the product outside the school. The group consists of three boys and two girls. The administration prefers a committee-selection arrangement that can be made in at most five different ways. If the number of possible selections exceeds five, the school administration will appoint the committee members instead.

Task:

Using mathematical justifications, Advise;

(a) The group.

(b) The administration

ITEM 4

A certain community in your village is experiencing increasing demand for clean water. To improve water supply, the district council plans to install solar panels to power a water pumping system.

During the planning , engineers established that:

- The product of the two possible numbers of solar panels that can effectively operate the system is 72.
- The sum of the two possible numbers of solar panels is 18.

The council needs to determine the possible numbers of solar panels that can be installed. The engineer advised that the project will only be considered possible if the required number of solar panels is at least 6 but not more than 15.

To meet future demand, the district plans to add water storage tanks annually. The number of tanks added each year doubles compared to the number added in the previous year.

In the first year, 4 tanks will be added. In the second year, 8 tanks will be added, followed by 16 tanks.

The council intends to continue the program until the cumulative number of tanks installed reaches 508. However, if the time required to achieve this target exceeds 5 years, the council will not provide funding for the upgrading of the existing water supply system.

To maximize solar energy production, council has purchased six solar panels labelled **A, B, C, D, E,** and **F**. Due to limited space on the mounting frame, only **four panels** can be installed in the first row. The order in which the panels are arranged affects the efficiency of power collection and maintenance operations.

The council therefore need to determine the number of different ways in which four panels can be arranged from the six available panels. If the number of possible arrangements exceeds **300**, a computer simulation will be used to identify the most efficient arrangement.

Task:

Based on mathematical validation, help the district council to make recommendations.

PART II

Respond to only one item in this part

ITEM 5

The canteen committee of a certain school recently introduced a new product. The administration is however not comfortable with the new product and has decide to carry out a survey. It got views from representatives of all the thirty streams in school. The learners were asked to rate the new product on a scale of 1-5 and the information summarized in the table below.

Scale	1	2	3	4	5	
No. Of Students	2	8	12	5	3	

If the variance is greater than 1, the administration will direct the committee to stop selling the new item.

The school administration further wants to compare the sales at the canteen over the past 9 terms as shown in the table below, on a quarterly basis.

TERM	1	2	3	4	5	6	7	8	9
SALES(In millions)	60	84	42	72	64	86	42	74	67

If the estimated sales in the **10th term** are greater than UGX. **70 million shillings,** the school will consider renewing the contract of the committee for two more years.

Additionally, the canteen is opened thrice a day. However the teacher in charge of meals has complained that most students have only 2 out of the four meals given by the school, leading to food wastage. He gives his finding below as proof of his claims.

x	1	2	3	4
P(X=x)	0.16	0.35	0.41	0.08

Where x represents the number of meals had, and P(X=x) represents the respective probabilities. If the expected number of meals had is less the 3, the canteen will be opened only twice a day.

Some teachers have also complained that students go late to class because they delay at the canteen. The probability that a learner is late for class is 0.25. A sample of 20 learners is investigated and If the probability of at least 8 of them are late is greater than 0.5, the school will recommend that the canteen opens only once a day.

Task:

Basing on numerical evidence, help the school administration to make appropriate decisions.

ITEM 6

An agricultural officer is evaluating the productivity of coffee farmers in a selected farming community in order to improve agricultural support programs. During a harvest season, the masses of coffee harvested were recorded as shown in the below.

MASS (KG)	10 – 14	15 – 19	20 – 24	25 – 29	30 – 34
NUMBER OF BAGS	4	6	10	12	8

The agricultural officer believes that the variation in production is manageable if the variance does not exceed 30.

The table below shows the amount of fertiliser used and the corresponding coffee yield obtained from selected farms.

Fertiliser Used (kg)	2	4	6	8	10	12
Coffee Yield (bags)	15	18	22	25	28	32

The agricultural officer will encourage appropriate fertiliser use if the estimated coffee yield, when 9 kg of fertiliser is used, exceeds 30 bags.

In the same community, the probability that a farmer uses improved seeds is 0.7, while the probability that a farmer uses irrigation is 0.5.

These events are considered to be independent. The agricultural officer will encourage farmers to adopt both practices in order to improve agricultural productivity if the probability that a farmer uses both practices does not exceed 0.40.

The office also recorded the daily amount of coffee (in tonnes) processed by a coffee factory as a continuous random variable, X, with probability density function:

$$f(x) = kx^2, \text{ for } 0 \leq x \leq 4$$

and

$$f(x) = 0, \text{ elsewhere.}$$

The agricultural officer will encourage the expansion of coffee processing activities if the expected daily amount of coffee processed exceeds 2.5 tonnes.

Task:
Use mathematical justifications to advise the officer.

END